Physics
Higher level
Markscheme
B.4 Thermodynamics

8 November 2026

W4ss Academy practice test 2 | **Topic** B.4 Thermodynamics | **Ref.** W4SS–PHY–B4–02

Student name

Paper 1A and Paper 2

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- Paper 1A: one mark per correct answer. The method (in French) shows how to find each answer.
- Paper 2: award one mark per ✓ in the markscheme. Accept equivalent answers (OWTTE).
- Total: **[20 marks]**.

Paper 1A — answers and methods

Question	Answer	Method
1.	**B**	1. Chaleur reçue pour fondre la glace : $Q = mL = 0.050 \times 335 = 16.75$ kJ. 2. La fusion se fait à 0 °C = 273 K : $\Delta S = Q / T = 16.75 / 273 = 0.061$ kJ K^{-1}.
2.	**A**	1. L'énergie interne est une fonction d'état : elle ne dépend que de l'état du gaz. 2. Après un cycle, le gaz revient à son état initial, donc $\Delta U = 0$.
3.	**A**	1. $S = k_B \ln \Omega$: l'entropie augmente avec le nombre de micro-états Ω. 2. Dans un système isolé, une transformation irréversible fait augmenter l'entropie : l'état final a **plus** de micro-états que l'état initial.

Paper 2 — markscheme

Question	Marks	Answers
1. (a)	**[2]**	***ALTERNATIVE 1*** «considering expansions from A» an adiabatic process will reduce/change temperature ✓ and so curve AC must be the steeper ✓ ***ALTERNATIVE 2*** temperature drop occurs for BC ✓ therefore CA must increase temperature «via adiabatic process». ✓
1. (b)	**[2]**	***ALTERNATIVE 1*** Use of adiabatic formula « $p_A V_A^{\frac{5}{3}} = p_C V_C^{\frac{5}{3}} \Rightarrow$ » $V_C = \frac{p_A}{p_C}^{\frac{3}{5}} V_A$ ✓ $V_C = \frac{5.00 \times 10^5}{4.60 \times 10^3}^{\frac{3}{5}} \times 2.00 \times 10^{-3}$ «$= 3.333 \times 10^{-2}\ m^3$» ✓ *For **MP2** , working or answer to at least 4 SF must be seen.* ***ALTERNATIVE 2*** $V_C = V_B$ ***AND*** $p_A V_A = p_B V_B$ ✓ $V_C = \frac{5 \times 10^5 \times 2 \times 10^{-3}}{3 \times 10^4}$ ✓ ***ALTERNATIVE 3*** $V_C = V_B$ ***AND*** $n = 0.2$ mol ✓ $V_C = (0.2 \times 8.31 \times 602) / 4 \times 10^4$ ✓
1. (c)	**[2]**	Increasing ✓ because thermal energy/heat is being provided to the gas « and temperature is constant, $\Delta S = \frac{\Delta Q}{T}$ ✓
1. (d)	**[2]**	***ALTERNATIVE 1*** $Q = \Delta U = \frac{3}{2} V_C \Delta P$ ✓ $Q =$ « $\frac{3}{2} \times 3.33 \times 10^{-2} \times (3.00 \times 10^4 - 4.60 \times 10^3)$» $= 1268.7 \approx 1270$ «J» ✓ ***Award [2]*** *for* ***BCA*** . *Accept negative values.* ***ALTERNATIVE 2*** $Rn = \frac{5 \times 10^5 \times 2 \times 10^{-3}}{602} = 1.66$ ***OR*** $T_c = 4.6 \times 10^3 \times 3.33 \times 10^{-2} \times 1.66 = 92.2$ ✓ $\Delta U = \frac{3}{2} \times 1.661 \times (602 - 92.21) = 1270$ «J» ✓ *Award **MP1** if $T_c = 92$ taken from **(e)***

1. (e)	**[2]**	$e_c = 1 - \frac{92}{602} = 0.847$ ✓ this engine has $e < e_c$ as it should ✓ ***Award [0]*** *if no calculation shown.*
2. (a)	**[3]**	Recognises that 4 heads and 6 tails also required ✓ Total number of microstates = 672 ✓ Fractional number = 672/1024 = 0.66 ✓ *Allow ecf for MP3* Recognises that 4 heads and 6 tails also required ✓ Total number of microstates = 672 ✓ Fractional number = 672/1024 = 0.66 ✓ *Allow ecf for MP3*
2. (b)	**[2]**	Two chances in 1024 so once every 512 throws ✓ 512 throws take 8.5 h so (a reasonable estimate is half way through) on average 4.3 h ✓
2. (c)	**[2]**	$\Delta S = k_B \ln 120 - \ln 1$ ✓ $= 4.8 k_B$ ✓